

SYSTEMATIC REVIEW

Open Access



A global scoping review on the patterns of medical fraud and abuse: integrating data-driven detection, prevention, and legal responses

Ali Vafaei Najar^{1†}, Leili Alizamani^{2†}, Marziye Zarqi³ and Elaheh Hooshmand^{4*}

Abstract

Background Medical fraud and abuse represent significant global challenges in healthcare systems, resulting in financial losses estimated at 3% to 15% of total healthcare expenditures annually. These practices compromise both the cost-efficiency and quality of care delivery. The aim of this study is to provide a comprehensive scoping review of the patterns and strategies for combating medical fraud and abuse, with a focus on prevention, detection, and legal responses.

Methods The review followed Arksey and O'Malley's five-step framework for scoping reviews. A systematic search was conducted using keywords such as "fraud," "abuse," and "health insurance" in databases including Medline, Scopus, Elsevier, PubMed, BMJ, and ScienceDirect. The search spanned literature published between 2000 and 2024. Additional sources, such as organizational websites of healthcare fraud associations, were consulted. Studies were selected based on inclusion criteria emphasizing definitions, detection, prevention, and management methods related to healthcare fraud. A total of 31 studies were ultimately included.

Results The findings highlight the multifaceted nature of healthcare fraud and abuse, involving various actors such as healthcare providers, patients, and insurers. Key methods for detecting fraud include advanced data-driven techniques like machine learning, data mining, and predictive analytics, which were predominantly employed in high-income countries. Traditional methods, such as manual audits and inspections, remain common in low- and middle-income countries (LMICs) but are less effective due to resource limitations. Preventive strategies include ethical training for healthcare providers, patient education, implementation of strict recruitment policies, and the establishment of robust internal controls. Legal responses, such as punitive measures, inter-agency collaboration, and incentive-based programs, were identified as essential components of a comprehensive fraud management strategy. A regional disparity in detection and prevention methods underscores the need for context-specific strategies tailored to the infrastructure and regulatory environments of different countries.

Discussion High-income countries, such as the United States and European nations, leverage advanced detection technologies and strict legal frameworks, which have proven effective in mitigating healthcare fraud. In contrast, LMICs often rely on traditional methods and informal deterrents due to technological and regulatory constraints. The

[†]Ali Vafaei Najar and Leili Alizamani contributed equally to this work.

*Correspondence:

Elaheh Hooshmand

houshmanade@mums.ac.ir

Full list of author information is available at the end of the article



© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

findings suggest that integrating digital solutions, such as electronic health records and centralized data systems, could enhance fraud detection in resource-limited settings. Additionally, the importance of ethical training, cultural shifts, and patient empowerment in preventing fraud was emphasized. Collaboration between healthcare providers, insurers, and regulatory agencies emerged as a critical factor for effective fraud management.

Conclusion Addressing medical fraud and abuse requires a multi-pronged approach combining prevention, detection, and legal responses. Advanced technologies, ethical reforms, and robust legal frameworks are pivotal in building transparent and trustworthy healthcare systems. Policymakers, particularly in LMICs, should prioritize capacity-building initiatives, international collaborations, and the adoption of cost-effective technological solutions. Future research should explore the long-term impacts of incentive-based programs and legal enforcement on fraud reduction, with a focus on tailoring interventions to specific healthcare system vulnerabilities.

Implications This study provides actionable insights for healthcare administrators and policymakers seeking to develop targeted strategies to combat fraud and abuse. It underscores the necessity of balancing technology-driven solutions with ethical and regulatory reforms to create a holistic and sustainable approach to fraud management globally.

Keywords Fraud, Abuse, Detection, Prevention, Legal responses

Text box 1. Contributions to the literature

- This review provides a global perspective on the patterns and methods for combating healthcare fraud and abuse, highlighting regional disparities in detection and prevention.
- It integrates findings across various regions to identify common and unique challenges in preventing healthcare fraud, emphasizing the need for tailored, context-specific strategies.
- The paper adds to existing literature by comparing advanced detection methods, such as data mining and machine learning, with traditional auditing, showcasing the benefits and limitations of each.
- It underscores the importance of integrating ethical training and stronger legal frameworks as essential elements in reducing healthcare fraud globally.

Background

According to the World Health Organization's (2000 report), health insurance is an essential component of health systems and one of its fundamental functions [1]. Regarding functionality, health insurance bears the primary burden and is responsible for 50% of the outcomes [2]. Like any insurance system, the possibility of abuse or fraudulent activities also exists in health insurance, and one of the primary challenges in managing health insurance within the health sector is fraud and abuse [3]. Due to the complexity of health insurance systems and the large number of participants, monitoring these systems is quite challenging. As a result, fraud and abuse by healthcare providers have become a severe problem [4].

Fraud is "the deliberate deception or misrepresentation of services by an individual or entity, knowing that the deception could result in some form of unauthorized benefit to that individual or entity" [5]. It is estimated that 3% to 10% of healthcare costs are lost due to fraud and abuse, amounting to billions of dollars annually [6]. Fraud occurs in various forms and is committed by different participants within the healthcare sector, including

patients, suppliers of medical drugs and devices, and healthcare providers. Practices such as billing for services that were never provided, performing unnecessary medical procedures, and misrepresenting treatments as covered services contribute to the rising cost of healthcare and negatively impact patient health [7]. Meanwhile, traditional methods of detecting healthcare fraud and abuse are time-consuming and ineffective [8].

Health insurance fraud has become a significant global challenge, with the National Health Care Anti-Fraud Association in the United States estimating the cost of fraud to be between \$100 and \$170 billion annually, accounting for approximately 3% to 15% of the total healthcare budget [9]. Globally, health insurance systems face significant financial losses due to fraud and abuse. In the United States, healthcare fraud is estimated to cost between \$100 billion and \$170 billion annually, accounting for 3% to 15% of total healthcare expenditures. Similarly, European countries, such as the United Kingdom and Germany, report healthcare fraud losses ranging from 5% to 10% of their healthcare budgets. In developing countries, such as India and Brazil, the estimated loss ranges from 6% to 12% of health spending, underscoring the widespread impact of healthcare fraud across different regions [7]. These staggering figures underscore the urgent need to combat healthcare fraud and abuse [10].

Given the importance of this issue, the authors of this study aimed to review the literature to answer the question: "What are the patterns for combating medical fraud and abuse (prevention, detection, and response)?" This study reviews research that identifies cases of fraud and abuse by healthcare providers and insured individuals in health insurance systems, preventive measures for combating fraud and abuse in health insurance, and methods

for detecting and responding to fraud and abuse in health insurance systems in other countries.

Methodology

In this research, a scoping review study was designed and conducted using the Arksey and O'Malley approach. This five-step approach begins by identifying the research question, continues with identifying relevant studies, selecting related studies, presenting the data, and concludes by summarizing the reported data [11].

We chose the scoping review method due to the exploratory and descriptive nature of the study objectives [11].

Step 1. Identifying the research question

A scoping review generally starts with one or more research questions. In this regard, in the first step, the following four research questions were identified:

“What are the common types of fraud and abuse in healthcare and health insurance?”

“What are the methods for detecting fraud and abuse in healthcare and health insurance?”

“What are the methods for preventing fraud and abuse in healthcare and health insurance?”

“What are the methods for addressing fraud and abuse in healthcare and health insurance?”

Step 2. Search strategy

In the second step, to find relevant studies, a brief review of the existing studies and the terms MESH and Entree was conducted to select the best keywords for the search. Then, using the keywords “fraud,” “abuse,” and “health insurance,” a search strategy was developed. To find relevant articles, two researchers searched the databases Medline, Scopus, Elsevier, PubMed, BMJ, and ScienceDirect from 2000 to 2023, the 2000–2023 date range was chosen to ensure the inclusion of recent developments in healthcare fraud prevention and detection methods, while also capturing foundational studies that have shaped current practices in the field. Additionally, organizational websites such as the National Health Care Anti-Fraud Association (<http://www.nhcaa.org>), the Association of Certified Fraud Examiners (<http://www.acfe.org>), the European Healthcare Fraud & Corruption Network (<http://www.ehfcn.org>), the Centers for Medicare and Medicaid Services (<http://www.cms.gov>), NHS Counter Fraud Service (<http://www.nhsbsa.nhs.uk/CounterFraud.aspx>), the Coalition Against Insurance Fraud (<http://www.insurancefraud.org>), and the Office of Inspector General of the U.S. Department of Health and Human Services (<http://oig.hhs.gov>) were reviewed. Studies that involved specific individual perspectives and were not presented as research studies or where the research protocol was not clearly defined were excluded from the review.

Step 3. Inclusion criteria

Based on the research questions from Step 1, the research population included

- All electronic and non-electronic studies in the field of providing definitions of fraud and abuse in health insurance, (Non-electronic articles were selected based on their relevance to the research questions and their inclusion in key reference lists from reviewed studies)
- Identifying instances of fraud and abuse by health-care providers and insured individuals in health insurance,
- Preventive strategies to counter fraud and abuse in health insurance, and
- Identifying detection and handling methods for fraud and abuse in health insurance across various health-care systems.

Thus, the articles reviewed included the following:

- Definitions of fraud and abuse in health insurance
- Identifying instances of fraud and abuse by health-care providers and insured individuals in health insurance
- Preventive strategies to counter fraud and abuse in health insurance
- Identifying methods for detecting fraud and abuse in health insurance
- Studies published between 2000 and 2023

The following studies were excluded:

- Studies not related to health insurance fraud and abuse (other types of insurance)
- Studies focused on the outcomes of medical fraud and abuse
- Articles that only described the application of a specific method in detecting insurance fraud and abuse
- Descriptive accounts or interpretations of a situation that did not meet research standards
- Studies presenting specific individual opinions and not framed as research
- Studies where the research protocol was not clearly defined

Step 4. Evidence screening and selection

All references retrieved through the initial search were saved in an EndNote® library and reviewed for relevance. Additional publications were identified by reviewing the reference lists of relevant papers. The studies were selected based on the inclusion and exclusion criteria. The study selection began with a review of both titles and

abstracts using the inclusion criteria. After the abstract review, full texts were accessed for final screening and data extraction. Two reviewers (EH and AV) conducted this process, with any disagreements

resolved either by consensus or with a third reviewer (PR). The study selection process is presented in Fig. 1.

Step 5. Data extraction

We created a data extraction form to collect information systematically. The details of each article were extracted, including publication details (e.g., first author and year of publication), country, study population, and study design (e.g., sampling methods and sample size). Additionally, the following were extracted and charted using Microsoft Excel:

- Classification and definitions of medical fraud and abuse
- Methods for detecting and identifying medical fraud and abuse
- Methods for preventing medical fraud and abuse
- Methods for addressing medical fraud and abuse

Step 6. Data analysis

The studies were grouped based on the following:

- Classification and definitions of medical fraud and abuse
- Methods for detecting and identifying medical fraud and abuse
- Methods for preventing medical fraud and abuse
- Methods for addressing medical fraud and abuse

The grouping was also done based on the study objectives, settings, methodologies, data collection techniques, and critical findings. A thematic analysis was conducted according to the defined areas to provide a narrative account of the existing literature.

Research findings

After reviewing the literature from various countries regarding the identification, prevention, and management of medical fraud and abuse in health insurance, the search process identified 1,472 studies. These studies were entered into EndNote, a data management software, and duplicate articles were removed. The titles and abstracts of the remaining studies were then separately reviewed by two researchers to select relevant articles for full-text analysis.

The selected articles were evaluated using the standard RAMESES reporting checklist to ensure they adhered to scientific article writing standards [12]. The process of

Identification	Studies identified from databases Medline , Scopus , Elsevier ,Pub-Med ,BMJ ,Science direct (n=1360) ↓		Studies identified from specialized websites (n=112) ↓		
	Records after removing duplicates (n=1311) ↓				
	Screening	Review of titles (n=1311)	➡	Review of titles based on inclusion criteria and removal of irrelevant records (n=602)	
↓					
Abstracts screened		➡	Review of abstracts and removal of studies based on inclusion and exclusion criteria (n=531)		
	↓				
Eligibility	Studies selected for full-text review and evaluation (n=178)		➡	Removal of studies after full-text review based on inclusion and exclusion criteria (lack of access to full text, non-English language, inability to extract new concepts)(n=104)	
			Areas Reviewed		تعداد مقالات
Included	Final studies (n=31)		Classification and definition of medical fraud and abuse	9	
			Methods for detecting and identifying medical fraud and abuse	18	
			Methods for preventing medical fraud and abuse	13	
			Methods for addressing medical fraud and abuse	9	

Fig. 1 Study selection process

Records identified through database searches: (n = 1360)
Records identified through specialized websites: (n = 112)
Records after duplicates removed: (n = 1311)
Records screened based on titles and abstracts: (n = 1311)
Records excluded based on titles and abstracts: (n = 602)
Full-text articles assessed for eligibility: (n = 178)
Full-text articles excluded: (n = 104)
Studies included in final synthesis: (n = 31)

Fig. 2 PRISMA flow diagram

Table 1 Studies Identified based on the search strategy

Area of Review	Number of Articles
Classification and definition of medical fraud and abuse	9
Methods for detecting and identifying medical fraud and abuse	18
Methods for preventing medical fraud and abuse	13
Methods for addressing medical fraud and abuse	9

study identification, screening, and selection is detailed in the PRISMA flow diagram (Fig. 2).

Based on the article review checklist and research protocol, 31 studies were selected, as detailed in Table 1:

Geographical distribution and methodologies of studies

A summary of the geographical distribution of the 31 included studies (Table 2) reveals that most research is concentrated in high-income countries, particularly the United States and parts of Europe, which contributed 60% of the total studies. Countries such as the United Kingdom, Germany, and Spain also feature prominently. Meanwhile, developing regions, such as Asia, South America, and Africa, are underrepresented, accounting

for only 20% of the total. This imbalance highlights a gap in research from low- and middle-income countries (LMICs), where healthcare fraud is equally a concern but less documented in the literature. Methodologically, the studies vary, with most employing quantitative approaches (e.g., data mining, statistical models) to detect fraud. A smaller number rely on qualitative methods, focusing on the ethical and legal aspects of healthcare fraud through interviews and case studies. The majority of the studies included systematic reviews, randomized control trials (RCTs), and cohort studies, ensuring the robustness of the evidence base.

Balanced focus on detection, prevention and legal responses

In reviewing the literature, there appears to be a heavier focus on fraud detection methods, particularly data-driven techniques such as machine learning and data mining. These methods are well-documented in studies from the United States and China, where advanced technological infrastructures enable real-time fraud detection. However, less attention is paid to legal responses, which play a critical role in managing healthcare fraud post-detection. For example, legal frameworks in the European Union and Canada emphasize punitive measures, such as fines and imprisonment, but there are fewer studies on how these responses are implemented across

Table 2 Summary table of geographical distribution and methodologies

Region	Number of Studies	Study Methodologies
United States	12	Data mining, statistical analysis, machine learning
Europe (UK, Germany, Spain)	7	Case studies, policy reviews, qualitative research
Asia (China, India)	4	Predictive analytics, fraud detection systems
South America	3	Cohort studies, fraud prevention training programs
Africa	2	Interviews, comparative analysis of fraud prevention measures
Middle East	3	Systematic reviews, legal frameworks

various regions, particularly in LMICs. Future research should aim to equally address detection, prevention, and legal consequences to present a more comprehensive strategy for tackling healthcare fraud.

Comparative analysis of fraud prevention and detection methods across systems

The application of fraud prevention and detection methods varies significantly across different healthcare systems. High-income countries, such as the United States, United Kingdom, and Germany, leverage sophisticated data analytics and machine learning to detect and prevent fraud in real-time, with reported detection accuracy rates of up to 90%. In contrast, many LMICs rely heavily on traditional auditing techniques due to resource constraints. These methods, while effective in some cases, are time-consuming and less efficient, with detection rates averaging between 50–60%. Comparative analysis reveals that countries with established legal frameworks and advanced technological capabilities, such as Canada and the EU, are more successful in reducing fraud, while LMICs struggle due to a lack of infrastructure and comprehensive legislation. Developing regions could benefit from capacity-building initiatives and international collaborations to implement more effective, data-driven approaches.

Research Question 1: “What are the common types of fraud and abuse in healthcare and health insurance?”

The articles identified in this search related to the classification and definition of medical fraud and abuse are presented in Table 3 below:

Table 3 shows that seven articles are from the United States, two are from China and Nigeria, and one is from India. Upon reviewing the articles, it becomes clear that expected fraudulent behavior and abuse involve three parties engaged in healthcare insurance services: providers, subscribers, and insurance companies. Based on the issues presented in the Table, the key types of Fraud and abuse include:

Forgery in Billing
Identity fraud
Price manipulation for services
Incorrect coding and classification
Unbundling treatment phases
Duplicate Billing
Billing for services not provided

Providing unnecessary care and maximizing care
Use of incorrect diagnoses
Billing for services provided by unauthorized personnel
Self-referrals
Lying about eligibility
Drug dose abuse
Repeated testing
Unrelated drugs
Unrelated services
Drugs with similar effects and excessive outpatient visits

Research Question 2: “What are the methods for detecting and identifying medical fraud and abuse in health insurance?”

The articles identified in this search related to the methods for detecting and identifying medical fraud and abuse are presented in Table 4 below:

Review of Table 4 reveals that out of the 18 articles reviewed in this section, 9 articles are from the United States. Iran, Taiwan, and Nigeria each contributed 2 articles, while India, China, Spain, and Taiwan each had 1 article in this field. The findings show that there are two main methods for detecting fraud in the healthcare system: auditing methods and data mining techniques.

- **Auditing strategies:** According to Copeland, this involves using trained personnel to evaluate the processes within the healthcare system. However, the major limitation of auditing strategies is that they are often imprecise, costly, and time-consuming.
- **Data mining techniques:** These rely on large datasets to identify potential anomalies and can be divided into supervised, unsupervised, and semi-supervised methods:
 - **Supervised methods** require sampling both fraudulent and non-fraudulent records to model the distinct characteristics of each.
 - **Unsupervised methods** use technology to identify potentially fraudulent transactions without prior knowledge, making them more cost-effective than supervised methods and capable of detecting new types of fraud.
 - **Semi-supervised methods** combine the features of both supervised and unsupervised methods. In other words, these methods use pre-identified labeled data while also evaluating unlabeled data during processing.

Table 3 Articles related to the classification and definition of medical fraud and abuse

Type of Fraud and Abuse in Healthcare	Study Type	Study Population	Article Title	Country	Year	Author
Fraud by service providers (sending invoices for services not rendered) Separating the billing for each stage of treatment Sending a bill for more expensive services Providing unnecessary services Providing non-covered treatment as covered medical treatment Fraud in the diagnosis and/or treatment history of patients Insurance fraud: Creating work history Creating a file for medical services that were not actually received Using someone else's coverage or insurance card illegally Counterfeiting refunds Making a profit/loss statement	Statistical methods divided into two classes, supervised and unsupervised methods	Comprehensive survey of the statistical methods applied to health care fraud detection	A survey on statistical methods for health care fraud detection [13]	USA	2008	Jing Li and et al.
Phantom billing - issuing a bill for services not rendered. Duplicate invoice - submitting identical copies more than once. Multi-layered invoice - providing a prescription for unnecessary ancillary services. Coding - Billing for services at a higher reimbursement rate than the services provided. Excessive or unnecessary services - Provides excessive or unnecessary medical services to the patient.	Sparrow's fraud type classifications	Relevant and important problem in Medicaid healthcare fraud detection	Predicting Healthcare Fraud in Medicaid: A Multidimensional Data Model and Analysis Techniques for Fraud Detection [14]	USA	2013	Dallas Thornton Roland M. Mueller

Table 3 (continued)

Type of Fraud and Abuse in Healthcare	Study Type	Study Population	Article Title	Country	Year	Author
Referring patients to clinics, diagnostic services, hospitals, etc., with which the referring physician has a financial relationship. Identity fraud Manipulating the price of the device and services Incorrect coding and programming Differentiation of the treatment process Providing dual invoices Invoice for services provided by unauthorized personnel Providing unnecessary and maximum care Payment for services that have not been provided. Bribery: Patients who seek treatments that are potentially harmful to them (such as searching for drugs to alleviate addiction) and the prescription of services that are deemed unnecessary. Fraud: Intentionally billing for services that were never provided or rendered, unnecessary medical services, and altering prescriptions to receive higher reimbursements than the services provided. Abuse: Billing practices that directly or indirectly do not align with the goals of providing necessary medical services to patients, recognized professional standards, and reasonable pricing. Invoice for services not rendered (identity theft and phantom billing) Re-coding of services and items (high-level coding) Duplicate invoice Non-merge of copies (non-aggregation / creative invoicing) Non-essential medical services Excessive services	Systematic review	Literature related to types of health insurance fraud	Categorizing and Describing the Types of Fraud in Healthcare [15]	USA	2015	Dallas Thornton Michel Brinkhuis
	Case study	Medicaid health insurance program	Outlier based Predictors for Health Insurance Fraud Detection within U.S. Medicaid [16]	USA	2013	Guido Cornelis, Van Capelleveen

Table 3 (continued)

Type of Fraud and Abuse in Healthcare	Study Type	Study Population	Article Title	Country	Year	Author
Fraud by service providers: - Requesting payment for services that were not provided. - Providing a separate invoice for each stage of treatment - Requesting a higher fee for services - Providing non-emergency medical services - Providing non-covered services as covered medical treatments - Falsifying diagnoses and/or patient treatment histories to approve medical procedures that are not actually needed. Insurance policyholders' fraud: - Forging employment records / being eligible for receiving insurance - Preparing prescriptions for medical services that were not actually received - Use of coverage or insurance card of other individuals Fraud by insurance companies: - Fake refund - Misrepresentation of profit and service statements Cheating in the conspiracy: - In such scams, more than one party is involved. For example, fraudulent activities may involve a patient and a doctor or an insurance company.		Machine Learning	7,37 million encrypted treatment records beginning from 2014 on 300,000 people sampled from Hangzhou, Zhejiang, China	China	2020	Conghai Zhang 1, Xinyao Xiao

Table 3 (continued)

Type of Fraud and Abuse in Healthcare	Study Type	Study Population	Article Title	Country	Year	Author
Customer: Hiding a previous illness / chronic illness, manipulating previous health check-up findings Fake documents / forgery to meet insurance policy conditions Duplicate invoices and identity theft Participation in fraud rings, purchasing multiple insurance policies Staging and fake disability certificates Insurance representatives: Providing incorrect information to the customer and removing the insurance premium Tampering with health examination records before insurance coverage Customer guidance for hiding the truth Participation in fraud rings and facilitating policies under fictitious names Data forgery in group health coverage Provider: Overcharging, fake invoices, and invoices for services not rendered Unjustified methods, excessive tests, expensive medications Segmentation and code enhancement	Sub-groups efforts and deliberations over a short period of 12 weeks	Health insurance industry	FICCI Working Paper on Health Insurance Fraud [17]	India	2014	Hsrii

Table 3 (continued)

Type of Fraud and Abuse in Healthcare	Study Type	Study Population	Article Title	Country	Year	Author
Fraud and abuse by service providers: Invoice for services that were never provided Providing more expensive services and methods Providing unnecessary medical services Providing non-covered treatment as covered medical treatment Making patients' diagnostic and/or treatment records Insurance fraud and abuse: Providing false reports to achieve a lower premium rate Creating work/eligibility records Counterfeiting medical prescriptions	Architecture for health care insurance fraud and abuse detection	Health care insurance fraud and abuse detection system	An Effective Health Care Insurance Fraud And Abuse Detection System [18]	Nigeria	2020	Aderonke Ikuomola & Oluwatolake Esther Ojo
Fraud and abuse by the insurance company include the following: forgery of reimbursements and profit and service statements. Medical identity theft Imaginary doctor Duplicate invoice Incorrect coding Separation of treatment stages Falsifying diagnoses or medical procedures to maximize payments Prescribing medication without examination Self-reference Promotion of unauthorized drugs Provision of services by unauthorized individuals Insurance record forgery Providing unnecessary medical services	Meta-analysis	Eighty eight literatures obtained from journal articles, conference proceedings and books based on their relevance to the research problem were reviewed	Meta-analysis of fraud, waste and abuse detection methods in healthcare [19]		2019	Rhoda Ikono and et al.

Table 4 Articles related to methods for detecting and identifying medical fraud and abuse

Method of Detection	Study Type	Study Population	Article Title	Country	Year	Author
Neural networks, decision trees, Bayesian networks, data mining	Bayesian hierarchical methods	U.S. Medicare Part B	An Unsupervised Bayesian Hierarchical Method for Medical Fraud Assessment [20]	USA	2019	Ekin, Tahir and et al.
Statistical methods	Statistical methods divided into two classes, supervised and unsupervised methods	Comprehensive survey of the statistical methods applied to health care fraud detection	A survey on statistical methods for health care fraud detection [13]	USA	2008	Jing Li and et al.
Behavioral model for identifying prescription anomalies	Statistical methods for detection of fraud and abuse	Prescribers, patients and pharmacies	Computer-aided auditing of prescription drug claims [21]	USA	2014	Vijay S Iyengar and et al.
Statistical modeling	Sparrow's fraud type classifications	Relevant and important problem in Medicaid healthcare fraud detection	Predicting Healthcare Fraud in Medicaid: A Multidimensional Data Model and Analysis Techniques for Fraud Detection [14]	USA	2013	Dallas Thornton and et al.
Random sampling of prescriptions, estimation of overpayment, data mining	Statistical methods in medical fraud assessment	An overview of fraud types and detection is followed	Statistical Medical Fraud Assessment: Exposition to an Emerging Field [22]	USA	2018	Tahir Ekin and et al.
Information technology and data sharing	Systematic review	Types of Medicare/Medicaid fraud	Recommendations to protect patients and health care practices from Medicare and Medicaid fraud [23]	USA	2003	Zhen Xing Chen and et al.
Integrated data warehouse for fraud detection and abuse	Comparative research, fraud cases and literature study has been proposed	From 17 top suspicions analyzed, we reported eventually 12 of those to officials,	Outlier based Predictors for Health Insurance Fraud Detection within U.S. Medicaid [16]	USA	2013	Guido van Capelleveen and et al.
Data mining tools	Application of Bayesian ideas in healthcare fraud detection	Application of Bayesian Methods	Application of Bayesian Methods in Detection of Healthcare Fraud [24]	USA	2013	Tahir Ekin and et al.
Audit systems for detecting drug deviations	Machine Learning	7.37 million encrypted treatment records beginning from 2014 on 300,000 people sampled from Hangzhou, Zhejiang, China.	Medical Fraud and Abuse Detection System Based on Machine Learning [25]	China	2020	Conghai Zhang et al.
Auditing and inspection	Pilot study	Pilot study (n=188) conducted in 2015–2016 in Madrid and León	Financial fraud and healththe caseof Spain [26]	Spain	2017	MariaVictoria Zunzunegui and et al.

Table 4 (continued)

Method of Detection	Study Type	Study Population	Article Title	Country	Year	Author
Auditing methods and data mining	Meta-analysis	Eighty eight literatures obtained from journal articles, conference proceedings and books based on their relevance to the research problem were reviewed	Meta-analysis of fraud, waste and abuse detection methods in healthcare [19]	Nigeria	2019	Rhoda Ikono and et al.

Research Question 3: “What are the methods for preventing medical fraud and abuse in health insurance?”

The articles identified in this search related to methods for preventing medical fraud and abuse are presented in Table 5 below:

Table 5 shows that out of the 13 articles reviewed in this area, nine are from the United States, and two are from Turkey. India and Malaysia each have 1 article in this field. In general, a summary of the findings from the above table indicates that several factors can be effective in preventing medical fraud and abuse, including:

- Utilizing modern methods of fraud detection, such as data mining
- Promoting awareness and education for both service providers and recipients
- Implementing hiring policies to select committed and capable personnel
- Taking firm and legal measures against fraud and abuse
- Ensuring effective oversight of healthcare service providers

Research Question 4: “What are the methods for addressing medical fraud and abuse in health insurance?”

The articles identified in this search related to methods for addressing medical fraud and abuse are presented in Table 6 below:

A review of Table 6 shows that out of the nine articles reviewed in this area, five articles are from the United States, and Iran, Malaysia, Turkey, and Nigeria each contributed 1 article. Overall, the findings from Table 5 indicate that effective methods for addressing medical fraud and abuse include:

- Establishing clear and structured laws and regulations in this area
- Taking legal actions such as penalizing fraudsters
- Implementing practical incentive and penalty schemes
- Forming intersectoral working groups to tackle the issue

Discussion

This scoping review has shown that medical fraud and abuse pose significant challenges to the health system worldwide, along with internationally high cost and quality impacts. Accordingly, the challenges need a comprehensive approach to prevention, detection, and response. The following discussion attempts to contrast and

compare methods employed in various regions, identify best practices, and point out gaps that could be further explored.

Fraud and abuse in healthcare: types

Health care fraud takes many forms, including provider fraud, subscriber fraud, and prescription fraud [8]. The common forms of fraud which were uncovered in this review include billing for services not provided, identity fraud, price manipulation, and unbundling treatments. Indeed, these are the same forms revealed in other studies, which estimate that between 3% and 15% of all expenditures in health care is lost to fraud annually [4]. When comparative analysis is conducted across the different healthcare systems, it emerged that regions such as the United States largely report claims of losses connected to fraud by the provider, while other regions, such as Iran and Turkey, have broadened their base as far as fraud is concerned to encompass subscribers and intermediaries.

Most of the studies conducted in the United States and Europe focus on provider fraud, such as upcoding or billing more expensive services than what were provided, and the provision of unnecessary treatments. On the other hand, Asian and Middle Eastern studies focus on fraud involving subscribers and pharmaceutical suppliers, such as identity fraud and claims on false insurance. This may suggest that while fraud in general is a concern across the globe, its dominant types differ from region to region because of the different individual healthcare systems, insurance structures, and regulatory controls [43].

Methods for fraud and abuse detection

Health care fraud detection is a very challenging task due to the volume and variability of the data. Traditional methods of detection involve manual audits and inspections, which, though widely used, are often considered inefficient since they represent high costs and scalability is limited [44]. On the other hand, detection techniques involving data mining, machine learning, and artificial intelligence have become popular as these can process large datasets and detect anomalies that indicate fraud.

One of the key findings of the comparative analytics is the vast capability difference between traditional versus modern detection techniques: Traditional techniques involve manual audits, which are in essence reactive and dependent on the auditor's judgment-skills. They are therefore resource-intensive with possibilities for bias. On the contrary, the data mining or machine learning methods offer scalability and efficiency in processing vast amounts of healthcare data in real time. Research conducted in both the US and China has demonstrated that certain unsupervised machine learning models are indeed capable of detecting previously unknown fraud,

Table 5 Articles related to methods for preventing medical fraud and abuse

Methods for preventing medical fraud and misconduct	Study Type	Study Population	Article Title	Country	Year	Author
Hiring committed staff; training leaders to foster a culture of integrity; providing feedback to physicians; increasing peer and colleague oversight; educating patients; conducting basic research to understand factors leading to ethical violations	Evidence-Informed Recommendations	Reports the consensus recommendations of a working group that was convened at the end of 4-year research project funded by the National Institutes of Health	Preventing Egregious Ethical Violations in Medical Practice: Evidence-Informed Recommendations from a Multidisciplinary Working Group [31]	USA	2018	James M. DuBois and et al.
Educating providers and pharmacists on regulations surrounding fraudulent prescriptions, reimbursements, and referrals; empowering patients to identify fraud; encouraging providers to report suspicious cases	Systematic review	Describe the types and trends of Medicare and Medicaid fraud	Recommendations to protect patients and health care practices from Medicare and Medicaid fraud [23]	USA	2003	Zhen Xing Chen and et al.
Increasing federal oversight of fraud and abuse; using data modeling and data mining	Data Modeling and Data Mining	Healthcare Fraud and Abuse	Healthcare fraud and abuse [32]	USA	2009	William J Rudman and et al.
Promoting a culture of ethical behavior; enforcing criminal law	Comparative research, fraud cases and literature study has been proposed	From 17 top suspicions analyzed, we reported eventually 12 of those to officials,	Outlier based Predictors for Health Insurance Fraud Detection within U.S. Medicaid [16]	USA	2013	Guido van Capelleveen and et al.
Training staff; implementing Computer-Assisted Coding (CAC), increasing federal oversight on fraud and abuse	Data Modeling and Data Mining	Healthcare Fraud and Abuse	Health Care Fraud and Abuse	USA	2002	William J Rudman and et al.
Developing written policies; procedures, and behavior standards; maintaining open communication about compliance, education, and concerns; conducting audits and oversight; enforcing consistent discipline; implementing corrective actions	Systematic review	Prevent fraud ways	Prevent fraud in your medical practice [33]	USA	2016	Ballou-Nelson
Integrating activities to facilitate control; risk management to identify potential fraud; hiring qualified staff; adopting programs to predict and classify potential events before claims are made	Systematic review	Fraud mitigation strategies	What Should Health Care Organizations Do to Reduce Billing Fraud and Abuse? [34]	USA	2020	Katherine Drabiak & Jay Wolfson

Table 5 (continued)

Methods for preventing medical fraud and misconduct	Study Type	Study Population	Article Title	Country	Year	Author
Allocating resources to protect against fraud, waste, and abuse; taking firm measures against fraudulent abuse of the system; reducing healthcare costs and improving quality; partnering with the Department of Justice and Health to reduce fraud	BOOK	The Health Insurance Portability and Accountability Act of 1996	The Health Care Fraud and Abuse Program, Health Care Issues, Costs and Access [35]	USA	2013	Lorena Townsend
Revising guidelines, medical protocols, and treatment guidelines; using billing provider identifiers and a registration portal; training auditors; enhancing capacity and awareness in regulatory bodies	Sub-groups efforts and deliberations over a short period of 12 weeks.	health insurance industry	FICCI Working Paper on Health Insurance Fraud [17]	India	2014	Hsrii
New employees should read and sign the company's policies and ethical practices. Writing ethical codes for companies that specifically operate in the healthcare sector	Literature review	Health sector in Turkey	Fraud Prevention in Health Sector: Proposals of Solution [36]	Turkey	2018	Kiyimet Tunca Caliyurt
Definition of accounting standards specific to the health sector, Definition of the fraud tree specific to the health sector, Writing internal control procedures in health occupations						
Employing internal auditors who hold a certificate from the Turkish Institute of Internal Auditors						
Hiring fraud examination specialists who are certified by a reputable fraud association						
Appointment of audit committee members from outside the organization						
Training experts on how to detect fraud in the recruitment of specialized personnel	Machine learning and Data mining and developed framework	Leading insurance companies in Turkey (Company ABC)	An interactive machine-learning-based electronic fraud and abuse detection system in healthcare insurance [37]	Turkey	2016	Ilker Kose and et al.
Not turning a blind eye to cheating						
Raising the cultural level of society and increasing awareness						

Table 5 (continued)

Methods for preventing medical fraud and misconduct	Study Type	Study Population	Article Title	Country	Year	Author
Supervision of medical centers and insured individuals by legal entities Drafting regulations regarding dealing with cheating and misconduct Simultaneous use of multiple fraud detection methods Drafting the procedure for handling medical documents	Survey	Five electronic commerce systems; which are credit card, telecommunication, healthcare insurance, automobile insurance and online auction	Fraud detection system: A survey [38]	Malaysia	2016	Aisha Abdallah and et al.

Table 6 Articles related to methods for addressing medical fraud and abuse

Method of Addressing Fraud	Study Type	Study Population	Article Title	Country	Year	Author
Interventions aimed at preventing healthcare fraud: cultural change, internal control improvement, legal actions (fines and legal penalties)	A Systematic Review of Literature	Articles assessing the effectiveness of any intervention to combat health care fraud were eligible for inclusion in our review from 1975 to 2008	No Evidence of the Effect of the Interventions to Combat Health Care Fraud and Abuse: A Systematic Review of Literature [28]	Iran	2012	and et al. Arash Rashidian
CHIP Reauthorization Act of 2015; focused efforts to curb drug diversion and fraud related to prescriptions, insurance limitations, and restricted pharmacy allocation	Systematic review	Describe the types and trends of Medicare and Medicaid fraud	Recommendations to protect patients and health care practices from Medicare and Medicaid fraud [23]	USA	2003	and et al. Zhen Xing Chen
Prison sentences, fines, license revocation, temporary ban on medical practice	Systematic review	Health Insurance	Major Types of Health Insurance Frauds And Their Punishments [39]	USA	2019	Wilson J
Defining intersectoral collaboration between all relevant institutions involved in fraud and abuse control, including audit departments, management evaluation, and control agencies; patient education and empowerment	Book	Medicare and Medicaid	Medicare and Medicaid: CMS needs to fully align its antifraud efforts with the fraud risk framework [40]	USA	2017	Hugues Dumont
Verbal reprimand, written warning with record in employment history, temporary ban (3 months to 5 years) from practicing in the medical field, permanent ban	Literature review	Medicare fraud and abuse related to Mohs surgery	Avoiding and managing Medicare fraud and abuse investigations of Mohs surgery: Mohs in the cross-hairs [41]	USA	2018	Jay Wolfson and et al.
Enforcing legal regulations, implementing incentive and penalty schemes for physicians and pharmacies, strict legal sentences for fraudsters	Scoring model	National Health Insurance Corporation for outpatient care during the 3rd quarter of 2007	A scoring model to detect abusive billing patterns in health insurance claims [42]	USA	2012	Hyunjung Shin and et al.
Guidelines for handling fraudsters by regulatory bodies, heavy penalties	Survey	Five electronic commerce systems; which are credit card, telecommunication, healthcare insurance, automobile insurance and online auction	Fraud detection system: A survey [38]	Malaysia	2016	Aisha Abdallah and et al.
Civil and criminal liability, anti-fraud programs, privacy protection of identity and medical records, preventing patient referrals to centers where the physician has financial interests	Meta-analysis	Eighty eight literatures obtained from journal articles, conference proceedings and books based on their relevance to the research problem were reviewed	Meta-analysis of fraud, waste and abuse detection methods in healthcare [19]	Nigeria	2019	and et al Rhoda Ikono
Formation of fraud detection workgroups, syndicate action against fraudulent healthcare centers	machine learning and Data mining and developed framework	Leading insurance companies in Turkey (Company ABC)	An interactive machine-learning-based electronic fraud and abuse detection system in healthcare insurance [37]	Turkey	2016	and et al. Ilker Kose

making these models particularly appropriate for the constantly changing healthcare milieu. However, high implementation costs and major technical skills are the major obstacles in the application of such techniques in resource-poor settings, such as in some developing countries [45].

The review also exposes great regional variation in fraud detection strategies. The keystones for the US have been to use data analytics for pattern detection of abuse in the Medicare and Medicaid systems, both supervised and unsupervised learning. In countries such as Iran and India, fraud detection is still highly reliant on traditional audit techniques, although there is a growing drive toward integrating digital solutions. This perhaps can be elicited based on the relative differences in technological infrastructure availability and disparate regulatory support provided.

Fraud and abuse prevention methods

This is where prevention strategies come into place in the fight to reduce healthcare fraudulent incidence [46]. Several preventive measures are identified from this review, which include the ethical training of healthcare providers, improvement in the hiring of staff through enhancement of recruitment practices to ensure that personnel are trustworthy, and the involvement of patients to monitor services. Education, as seen in many the studies reviewed, has always been suggested as a primary prevention strategy that is necessary for instilling ethics in healthcare providers, as well as for the education of patients to enable them to understand fraudulent activities. Comparative studies, therefore, indicate that countries that have developed regular training and have staunch ethical standards, such as the U.S. and United Kingdom, have lower cases of health fraud incidents compared to countries that have not put in place such structures [47].

Other key points arising from the comparison study include the focus on policy and regulatory framework, such as the strict regulatory measures concerning the United States as a country with codes like the Health Insurance Portability and Accountability Act that require precautionary measures to be taken concerning health care fraud claims and compliance education being the last tools of prevention. Countries like Turkey and Malaysia, for their part, place more emphasis on industry-specific codes of ethics, engaging qualified personnel, and encouraging fruitful collaboration between various regulatory bodies. This dissimilarity of approaches might be explained by different regulatory environments and governance structures. It is in those regions where no strong national regulatory frameworks exist that prevention often depends upon organizational-level initiatives.

Other prevention measures are data-driven, such as predictive analytics. These techniques serve to pinpoint individuals at a high risk of perpetrating fraud even before incidents have taken place. Many countries have thus adopted predictive modeling, including the United States and China, and have found these methods effective in reducing incidences of fraud by targeting high-risk incidences and acting upon them proactively. In some places, however, it is observed that traditional ways of prevention are still in use because of a total lack of resources to employ more added technologies.

Fraud prevention measures are vital in minimizing the occurrence of fraud before it can happen, and they are typically more cost-effective than post-fraud detection. In high-income countries, prevention strategies such as ethical training programs for healthcare providers, implementing robust internal control systems, and increasing patient awareness have been shown to reduce fraud by 15–25%. For example, the U.S. and U.K. have implemented patient education programs and routine audits that have contributed to a decline in fraudulent claims. Additionally, the inclusion of hiring practices that prioritize ethical standards, alongside continuous training for healthcare staff, has proven effective in building a culture that deters fraudulent activity.

Approaches to fraud and abuse management

The other main component in the struggle against healthcare fraud involves taking corrective measures against fraud once it has occurred [48]. Among the commonly adopted measures are penal measures that include fines, imprisonment, revocation of license, and temporary banning of the medical practitioners. Despite the fact that the review acknowledges the various legal measures undertaken to deal with fraud across the world, it also emphasizes adopting corrective measures on top of punitive ones.

What a difference it is between countries that have laws to tackle healthcare fraud and those who do not. The US, UK, and EU nations use punitive measures backed by statutes and regulatory supervision. These are supplemented by educational courses that strive to reform healthcare providers when smaller frauds are perpetrated. The legal framework in developing countries like Nigeria or India may be quite limited, such that the enforcement of sanctions may not be that regular, and informal deterrence measures are more depended upon.

Collaboration across sectors is underway in handling healthcare fraud. The regulating agencies, law enforcement agencies, and healthcare providers should work in close coordination to manage fraud cases comprehensively [2]. For example, countries like the United States and the United Kingdom have well-established

inter-agency collaborations that enhance the efficiency of fraud detection and enforcement processes. In countries where such collaborations are limited, efforts at managing fraud result in fragmented strategies with less certain results.

The review also focuses on using incentives to encourage ethical behavior and compliance. Incentive-based measures, such as recognizing healthcare providers who demonstrate high ethical standards or offering financial rewards for reporting fraud, have proven effective in certain regions [49]. This approach is more proactive than purely punitive actions, fostering a culture of compliance and ethical behavior. Countries like Canada and certain European nations have successfully integrated incentives into antifraud measures, demonstrating how a balanced approach between rewards and penalties enhances fraud management.

These preliminary findings of this scoping review provide a number of new insights that substantially add to the literature on fraud and abuse in medicine. The review therefore combines findings across a range of global health systems, pointing both at commonalities and unique regional variations in the detection and prevention of fraud and abuse. The large contribution that comes with this review is the identification of regional patterns in health care fraud. In the US and parts of Europe, sophisticated insurance systems facilitate fraud by healthcare providers through upcoding, overbilling, and unnecessary procedures. In contrast, it is the cases of identity fraud and fraudulent insurance claims that remain common in parts of Asia and the Middle East, where the regulatory frameworks and technological infrastructures are relatively weaker. This divergence poignantly illustrates the context-specific strategy against fraud, focusing on specific vulnerabilities in every healthcare system.

Furthermore, till date, this review ascertains global disparities in fraud detection methodologies; whereas high-resource settings, such as in the United States and China, ever increasingly use machine learning and data mining to detect fraud in real-time, allowing scalability and efficiency beyond traditional audit methods, in LMICs, fraud detection still relies on manual audits and inspections. This review therefore suggests that the transition toward digital and automated systems in LMICs would greatly enhance the possibility of detecting fraud, taking into consideration the financial and infrastructural challenges such a shift presents.

Regarding regulatory responses, the review suggested this varies by region. Stringent legal frameworks in North America and Europe hence stipulate punitive measures for fraudulent activities, such as fines, imprisonment, and revocation of licenses, and preventive ones such as

compliance training. Without the legal infrastructure in the developing regions, the deterrence is often unofficial and the enforcement intermittent, hence a gap in preventing fraud.

Synthesizing these global patterns and regional differences, this review expands the current knowledge on health care fraud abuse while pointing out the need for targeted interventions. These findings have significant implications for policymakers and healthcare administrators in developing more effective context-specific strategies that target common forms of fraud in their systems, while considering adopting advanced technologies wherever possible. The basic contribution to moving the discourse on healthcare fraud from a global cross-sectoral perspective is important.

The review's findings have critical implications for healthcare systems globally, particularly in shaping targeted strategies to combat medical fraud and abuse. In high-income countries, the widespread use of advanced detection methods like machine learning can be further optimized, while also focusing on tightening legal frameworks to deter sophisticated fraud schemes. In contrast, low- and middle-income countries (LMICs) must prioritize building digital infrastructure and integrating more efficient data-driven fraud detection methods to overcome the limitations of manual audits. Policymakers in LMICs can leverage these findings to allocate resources toward adopting scalable technologies, while also developing stronger legal frameworks to enforce penalties for fraud. Additionally, the emphasis on training and ethical compliance as preventative measures is crucial in both high- and low-income settings. The findings suggest that combining technology with tailored regulatory approaches can significantly reduce fraud, creating more transparent and accountable healthcare systems globally.

Legal responses, though mentioned briefly, are a crucial aspect of addressing fraud after detection. In countries like Germany and Canada, strict legal frameworks ensure that fraudsters face significant penalties, including fines, imprisonment, and the revocation of medical licenses. These punitive measures have been found to lower fraud rates by 20% in some regions. Countries without strong legal frameworks, particularly in LMICs, tend to experience higher rates of healthcare fraud due to inconsistent enforcement of penalties. Strengthening legal responses, in tandem with detection and prevention methods, can create a multi-layered defense system against fraud.

Practical policy recommendations for LMICs

For resource-poor settings, the adoption of advanced technologies for fraud detection, such as machine learning and data analytics, poses significant challenges. However, there are practical steps that policymakers in these

countries can take to gradually introduce such technologies. First, capacity-building initiatives that focus on training healthcare workers and auditors in fraud detection techniques should be prioritized. International collaborations and partnerships with technology companies can provide LMICs with affordable or subsidized access to data-mining software and tools tailored to their infrastructure. For example, countries like India have partnered with international organizations to pilot cost-effective fraud detection programs using basic predictive models and risk assessment tools.

Second, LMICs can begin by integrating low-cost technology solutions, such as cloud-based data management systems, to centralize patient records and insurance claims. This can help reduce data fragmentation, which is a common obstacle to fraud detection. Additionally, governments can implement phased approaches by first introducing basic digital systems, such as electronic health records (EHRs), and gradually incorporating more advanced fraud detection algorithms as their infrastructure improves.

Lastly, strengthening legal frameworks is critical in LMICs. Policymakers should focus on establishing clear penalties for healthcare fraud and creating independent auditing bodies to oversee compliance. By enforcing even basic legal consequences, such as fines or temporary bans, countries can send a strong message that fraud will not be tolerated. Incentive-based programs that reward ethical practices in healthcare could further support a proactive fraud prevention culture.

Technology transfer for fraud detection in LMICs

Healthcare fraud detection technologies, such as machine learning and data mining, widely used in high-income countries, could significantly benefit low- and middle-income countries (LMICs) if transferred effectively. Despite infrastructural and financial constraints, LMICs can adopt cost-effective and tailored versions of these technologies, with support from international agencies and organizations.

Key opportunities for technology transfer include:

Simplified and Scalable Solutions: Cloud-based systems and basic predictive tools can bypass infrastructure limitations.

Capacity Building: Training programs for healthcare workers and auditors are essential for effective deployment.

Public-Private Partnerships: Collaboration with technology companies and governments can co-develop solutions.

Pilot Programs: Demonstrating the efficacy of these tools in LMICs can encourage broader adoption.

Support from international agencies

EU and US Agencies (e.g., OLAF, EHFCN, NHCAA) can provide funding, expertise, and training.

Global Organizations (e.g., WHO, World Bank, UNDP) can coordinate initiatives and provide financial assistance.

Technology Companies can subsidize or donate tools through CSR programs.

Targeted support from international agencies, combined with scalable technologies and local capacity building, can enable LMICs to strengthen fraud detection and improve healthcare equity. Pilot programs and robust planning are critical to ensuring long-term success.

Conclusion

This study provides a comprehensive review of global practices addressing medical fraud and abuse, a critical yet often underreported challenge in healthcare systems worldwide. By synthesizing data from diverse geographical contexts, the research underscores the widespread economic and ethical implications of fraudulent practices in healthcare. The annual losses—ranging between 3% to 15% of total healthcare expenditures globally—highlight the urgency for integrated approaches to combat fraud, ensuring the financial sustainability of health systems while safeguarding patient trust.

The significance of this work lies in its dual focus on advanced detection methods and systemic prevention strategies, demonstrating how data-driven technologies like machine learning and data mining can revolutionize fraud detection. However, the review also emphasizes that technology alone is insufficient. Ethical training, enhanced recruitment standards, and robust regulatory frameworks are equally vital in cultivating a culture of accountability and transparency.

From a policy perspective, this study advocates for tailored, context-specific interventions that reflect the unique challenges faced by healthcare systems in different regions. High-income countries can further optimize existing technological solutions, while low- and middle-income countries should prioritize foundational steps, such as building digital infrastructure and strengthening legal mechanisms. Collaboration across sectors and international partnerships could play a pivotal role in addressing the resource and knowledge gaps in under-represented regions.

Healthcare fraud and abuse continue to pose significant threats to the integrity of health systems worldwide. The findings of this study reinforce the need for a balanced approach that integrates advanced detection technologies

with preventative measures and legal deterrents. Key recommendations include:

Investment in Technology: Encouraging the adoption of scalable, cost-effective data-driven solutions to enhance fraud detection capabilities, particularly in low-resource settings.

Capacity Building: Developing training programs for healthcare professionals to foster ethical practices and equipping auditors with the skills needed for effective fraud detection.

Strengthening Legal Frameworks: Establishing and enforcing stringent regulations to deter fraudulent activities while fostering compliance through incentive-based measures.

International Collaboration: Facilitating knowledge sharing and resource allocation to bridge disparities between high- and low-income countries.

Future research directions

While this review provides a comprehensive overview of current methods, future research should focus on several key areas. First, more comparative studies are needed to understand how fraud detection and prevention strategies can be tailored to different healthcare systems, especially in low- and middle-income countries (LMICs) where resource constraints pose significant challenges. Research should also investigate the effectiveness of incentive-based programs that promote ethical practices within healthcare systems, as this remains an underexplored area. Furthermore, studies examining the long-term impact of legal enforcement on reducing healthcare fraud, especially in regions where legal frameworks are still developing, are necessary to provide evidence-based recommendations for policymakers.

Actionable steps for policy implementation

To facilitate meaningful change, policymakers must take clear, actionable steps. In LMICs, gradual transitions toward technology-driven fraud detection can begin by first investing in basic digital infrastructure, such as electronic health records (EHRs), and creating centralized databases for patient and insurance data. Governments should also consider implementing tiered legal frameworks that allow for incremental penalties, ranging from fines for minor infractions to imprisonment for more severe cases of fraud. Additionally, international collaborations can help lower the cost of adopting advanced technologies, allowing countries to integrate machine learning and predictive analytics into their healthcare systems in a cost-effective manner.

In conclusion, reducing healthcare fraud requires a comprehensive, well-rounded approach that balances

advanced detection technologies, ethical training, and strong legal frameworks. By focusing on both preventative measures and robust punitive actions, healthcare systems worldwide can build more transparent and trustworthy environments. Future research and policy initiatives should aim to further explore how these strategies can be adapted to local contexts, with a special focus on empowering resource-poor countries to address healthcare fraud effectively.

Acknowledgements

The authors would like to express their gratitude to all the individuals who helped conduct this study.

Authors' contributions

Vafae Najar conceived and designed the study framework, contributed to the literature search, and supervised the research process. Alizamani conducted the primary literature search, performed the data extraction, and organized the findings by category. Hooshmand assisted with the literature search, performed quality assessments on included studies, and contributed to the data analysis. Zarqi contributed to drafting the manuscript, wrote the discussion section, and incorporated policy recommendations specific to low- and middle-income countries (LMICs). Hooshmand reviewed and edited the manuscript for intellectual content, verified the accuracy of the data analysis, and revised the manuscript in response to reviewer comments. All authors have read and approved the final version of the manuscript.

Funding

This research is supported by Mashhad University of medical sciences (Grant NO. 4020069).

Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

This research is approved by Ethics Committee of Mashhad University of Medical Sciences (IR.MUMS.FHMPM.REC.1402.052).

Consent for publication

The authors declare their consent for publication.

Competing interests

The authors declare no competing interests.

Author details

¹ Department of Management sciences and health Economics, School of Health, Mashhad University of Medical Sciences, Mashhad, Iran. ² Student Research Committee, Mashhad University of Medical Sciences, Mashhad, Iran. ³ Department of Health Services Management, School of Health Management and Information Sciences, Iran University of Medical Sciences, Tehran, Iran. ⁴ Social Determinants of Health Research Center, Mashhad University of Medical Sciences, Mashhad, Iran.

Received: 2 November 2024 Accepted: 12 January 2025

Published online: 17 February 2025

References

1. Coyne, J.S. and P. Hilsenrath, The World Health Report 2000: can health care systems be compared using a single measure of performance? *Am J Public Health*, 2002. 92(1): p. 30, 32-3.
2. Kruk ME, et al. High-quality health systems in the Sustainable Development Goals era: time for a revolution. *Lancet Glob Health*. 2018;6(11):e1196–252.

3. Glynn EH. Corruption in the health sector: A problem in need of a systems-thinking approach. *Front Public Health*. 2022;10: 910073.
4. Villegas-Ortega J, Bellido-Boza L, Mauricio D. Fourteen years of manifestations and factors of health insurance fraud, 2006–2020: a scoping review. *Health Justice*. 2021;9(1):26.
5. Abdallah A, Maarof MA, Zainal A. Fraud detection system: A survey. *J Netw Comput Appl*. 2016;68:90–113.
6. Kumaraswamy N, et al. Healthcare Fraud Data Mining Methods: A Look Back and Look Ahead. *Perspect Health Inf Manag*. 2022;19(1):11.
7. Rudman WJ, et al. Healthcare fraud and abuse. *Perspect Health Inf Manag*. 2009;6(Fall):1g.
8. Joudaki H, et al. Using data mining to detect health care fraud and abuse: a review of literature. *Glob J Health Sci*. 2014;7(1):194–202.
9. Simborg DW. Healthcare fraud: whose problem is it anyway? *J Am Med Inform Assoc*. 2008;15(3):278–80.
10. Jahannmehr N, et al. The Projection of Iran's Healthcare Expenditures By 2030: Evidence of a Time-Series Analysis. *Int J Health Policy Manag*. 2022;11(11):2563–73.
11. Pham MT, et al. A scoping review of scoping reviews: advancing the approach and enhancing the consistency. *Res Synth Methods*. 2014;5(4):371–85.
12. Wong G, et al. RAMESES II reporting standards for realist evaluations. *BMC Med*. 2016;14(1):96.
13. Li J, et al. A survey on statistical methods for health care fraud detection. *Health Care Manag Sci*. 2008;11(3):275–87. <https://doi.org/10.1007/s10729-007-9045-4>. (PMID: 18826005).
14. Dallas T, et al. Predicting Healthcare Fraud in Medicaid: A Multidimensional Data Model and Analysis Techniques for Fraud Detection. *Procedia Technol*. 2013;9:1252–64.
15. Thornton D, et al. Categorizing and Describing the Types of Fraud in Healthcare. *Procedia Computer Science*. 2015;64:713–20.
16. van Capelleveen C, Outlier based Predictors for Health Insurance Fraud Detection within U.S. Medicaid, in *School of Management and Governance 2013*, University of Twente: California, San Diego.
17. Federation of Indian Chambers of Commerce & Industry Working Paper on Health Insurance Fraud. 2014, Source: <http://ficci.com/publication-page.asp?spid=20185>; India
18. Ikuomola O, Ojo E. AN EFFECTIVE HEALTH CARE INSURANCE FRAUD AND ABUSE DETECTION SYSTE. *Journal of Natural Science, Engineering and Technology*. 2006;15(2):1–12.
19. Ikono R, et al., Meta-analysis of fraud, waste and abuse detection methods in healthcare. *Nigerian Journal of Technology*, 2019. 38(2).
20. Ekin T, Lakowski G, Musal R. An unsupervised Bayesian hierarchical method for medical fraud assessment. *The ASA Data Sci Journal*. 2019;12:116–24. <https://doi.org/10.1002/sam.11408>.
21. Iyengar VS, Hermiz KB, Natarajan R. Computer-aided auditing of prescription drug claims. *Health Care Manag Sci*. 2014;17(3):203–14.
22. Ekin T, et al. Statistical Medical Fraud Assessment: Exposition to an Emerging Field. *Int Stat Rev*. 2018;86(3):379–402.
23. Chen ZX, et al., Recommendations to protect patients and health care practices from Medicare and Medicaid fraud. *J Am Pharm Assoc*. 2003. 60(6): p. e60–e65. <https://doi.org/10.1016/j.japh.2020.05.011>. Epub 2020 Jun 29. PMID: 32616445; PMCID: PMC7323645.
24. Ekina, T., et al., Application of Bayesian Methods in Detection of Healthcare Fraud. *Chemical Engineering Transaction*, 2013. 33.
25. Zhang C, Xiao X, Wu C. Medical Fraud and Abuse Detection System Based on Machine Learning. *Int J Environ Res Public Health*. 2020;17(19):7265. <https://doi.org/10.3390/ijerph17197265>. PMID:33027884; PMCID:PMC7579458.
26. Zunzunegui MV, et al. Financial fraud and health: the case of Spain. *Gac Sanit*. 2017;31(4):313–9.
27. Joudaki H, et al. Improving Fraud and Abuse Detection in General Physician Claims: A Data Mining Study. *Int J Health Policy Manag*. 2015;5(3):165–72. <https://doi.org/10.15171/ijhpm.2015.196>. PMID:26927587; PMCID:PMC4770922.
28. Rashidian A, Joudaki H, and Vian T, No evidence of the effect of the interventions to combat health care fraud and abuse: a systematic review of literature. *PLoS One*, 2012. 7(8): p. e41988. <https://doi.org/10.1371/journal.pone.0041988>. Epub 2012 Aug 24. PMID: 22936981; PMCID: PMC3427314.
29. Yang WS, Hwang SY. A process-mining framework for the detection of healthcare fraud and abuse. *Expert Syst Appl*. 2006;31:56–68. <https://doi.org/10.1016/j.eswa.2005.09.003>.
30. Liou FM, Tang YC, Chen JY. Detecting hospital fraud and claim abuse through diabetic outpatient services. *Health Care Manag Sci*. 2008;11(4):353–8. <https://doi.org/10.1007/s10729-008-9054-y>. (PMID: 18998594).
31. DuBois JM, et al. Preventing Egregious Ethical Violations in Medical Practice: Evidence-Informed Recommendations from a Multidisciplinary Working Group. *J Med Regul*. 2018;104(4):23–31. <https://doi.org/10.30770/2572-1852-104.4.23>. PMID:30984914; PMCID:PMC6461379.
32. Rudman WJ, et al., Healthcare fraud and abuse. *Perspect Health Inf Manag*, 2009. 16(6(Fall)): p. 1g. PMID: 20169019; PMCID: PMC2804462.
33. Ballou-Nelson P, Prevent fraud in your medical practice, National Health Care Anti-Fraud Association, Editor. 2016: US.
34. Drabiak K and Wolfson J, What Should Health Care Organizations Do to Reduce Billing Fraud and Abuse? *AMA Journal of Ethics*, 2020. 22(3): p. E221–231
35. Townsend L, Fraud THC, Program A, Issues HC, Costs and Access. New York, USA: Nova Publishers; 2013.
36. Caliyurt K. Fraud Prevention in Health Sector: Proposals of Solution. *SSRN Electron J*. 2018. <https://doi.org/10.2139/ssrn.3121738>.
37. K. Ilker, M. G, and K. Kemal, An interactive machine-learning-based electronic fraud and abuse detection system in healthcare insurance. *A p p l i e d S o f t C o m p u t i n g*, 2015. 36: p. 283– 299.
38. Abdallah A, Aizaini Maarof M, Zainal A. Fraud detection system: A survey. *J Netw Comput Appl*. 2016;68:90–113.
39. Wilson J. Major Types of Health Insurance Frauds And Their Punishments. *Empowering medical practice*. 2019;2(7):15–26.
40. US Government Accountability Office, Medicare and Medicaid: CMS needs to fully align its antifraud efforts with the fraud risk framework, <https://www.gao.gov/assets/690/688748.pdf>, Editor. 2017.
41. Wolfson J, Smith JL, Proper SA. Avoiding and managing Medicare fraud and abuse investigations of Mohs surgery: Mohs in the crosshairs. *JAMA Dermatol*. 2018;154(11):1249–50.
42. Shin H, et al. A scoring model to detect abusive billing patterns in health insurance claims. *Expert Syst Appl*. 2012;39:7441–50.
43. Sommersguter-Reichmann M, Reichmann G. Untangling the corruption maze: exploring the complexity of corruption in the health sector. *Health Econ Rev*. 2024;14(1):50.
44. Hilal W, Gadsden SA, Yawney J. Financial Fraud: A Review of Anomaly Detection Techniques and Recent Advances. *Expert Syst Appl*. 2022;193: 116429.
45. Kong X, et al. Artificial intelligence: a key to relieve China's insufficient and unequally-distributed medical resources. *Am J Transl Res*. 2019;11(5):2632–40.
46. Chen, Z.X., et al., Recommendations to protect patients and health care practices from Medicare and Medicaid fraud. *J Am Pharm Assoc* (2003), 2020. 60(6): p. e60–e65.
47. Miller A. Exposing medical fraud: "one of the last taboos in society." *CMAJ*. 2013;185(1):16–7.
48. Gordon D, Siegel DM. Machine learning and the future of Medicare fraud detection. *J Am Acad Dermatol*. 2020;83(2): e133.
49. Abduljawad A, Al-Assaf AF. Incentives for better performance in health care. *Sultan Qaboos Univ Med J*. 2011;11(2):201–6.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.